Operator script: arrival

Team only | Read quoted text aloud | 9 September 2026 | v1.1

Use this pack

Give participants only the participant handout. Keep this operator script and the controller mapping out of their view. Use the dashboard's assigned order. These materials explain the existing study; they do not authorize robot motion or change the physical release requirements.

Use the university's existing information and consent process. Honour the recorded video choice; if the protocol cannot run with that choice, resolve this before recording. Do not promise that coded video is anonymous.

Welcome and intake

> Thanks for coming. We are studying a guided panel-handling task with a robot. We are evaluating the system, not your ability. Please read the study information, and ask us about anything that is unclear before deciding whether to take part.

After consent, assign the next unused study code and show the intake QR. Say:

> Please scan this code and check that the participant code on the form matches the screen. Answer the background questions based on your own experience. Your responses do not need to please the team. Tell us when you have submitted it.

Check receipt in the dashboard. If tracking is unavailable, check the confirmation screen and the matching Sheet row; record the fallback. Do not treat a tick or a verbal report alone as verified receipt.

Briefing after intake, before rehearsal

> You will complete nine trials, arranged as three blocks called A, B and C. Each block has three trials. We will use the same panel task throughout, and explain each trial before it starts.
>
> First, you will take the panel to the low gripper and keep supporting it until I tell you to let go. Then return to the start marker. During the lift you will either wait there or perform the one movement we demonstrate, when I say NOW. You will not need to guess what to do.
>
> Once the robot is stationary, I will tell you when to approach for the panel task. Afterwards, return to the marker and wait while the robot lowers the panel. Approach and support it only when instructed. Do not try to catch a falling panel.
>
> We will help fit the motion-capture equipment. Please tell us about discomfort or restricted movement. You can say STOP, ask for a break or end your participation at any time. We will show you the stop procedure and waiting area before starting.
>
> You will complete a short form after each three-trial block and one final form. Please answer about your own experience. We will explain the comparison after your final answers. What would you like us to clarify?

Demonstrate and check understanding

With robot motion stopped, point out the start marker, panel handoff, permitted task area and waiting area. Demonstrate the agreed panel task, both approved movements and the stop procedure. Use the same rehearsal procedure for everyone. Any physical demonstration must stay within the released procedure.

> Please show or tell me where you wait during the lift, what tells you to perform the movement, and what you do if you want to stop.

Correct misunderstandings neutrally before starting. Record any extra rehearsal or difficulty. Do not describe any controller or movement as the safer, smarter or better one.

Operator cue sheet: trials

Team only | Use one assigned scenario per trial | v1.1

Before each block

Read the block letter assigned in the dashboard in place of “A” below:

> This is Block A. There are three trials in this block. I will tell you what to do before each one. Afterwards, you will answer one short form about the whole block.

Confirm the participant code, assigned slot, recording filenames and real recording state. Keep controller names and model confidence off the participant-facing screen. Use the dashboard for experimenter controls; do not advance past an unmet physical requirement.

Brief the assigned lift action

Clean / baseline:

> During the lift, remain at the start marker. There will be no NOW cue in this trial. Wait until I invite you to approach the panel.

Distractor / non-closing movement:

> During the lift, on NOW, perform the non-closing movement we just practised once. Follow the demonstrated route, return to the start marker and wait.

Rapid intrusion / controlled closing movement:

> During the lift, on NOW, perform the closing movement we just practised once, within the demonstrated limit, then immediately return to the start marker and wait.

Use the already agreed direction, pace and endpoint. Do not invent those details while reading these cues. Keep the spoken NOW synchronized with the lift/event start specified in the protocol. Record cue time and measured movement onset separately; a cue is not proof of movement.

Read as the trial advances

1. **Ready:** “Stand at the start marker with the panel and wait.”
2. **Approach:** “Bring the panel to the low gripper and hold it as demonstrated.”
3. **Grip:** “Keep supporting the panel while we check the grip.” Check both cups before giving the next instruction.
4. **Retreat:** “You can let go now. Return fully to the start marker.” Confirm clearance before the deliberate lift confirmation.
5. **Lift:** For the two movement scenarios only, say “NOW” at the agreed lift/event start. For baseline, give no movement cue. Do not repeat a missed action during the same trial; record the deviation.
6. **Raised task:** After verifying the robot is stationary: “You may approach now. Complete the panel task exactly as demonstrated.” No second movement event occurs here.
7. **Return:** At the agreed task completion: “Return fully to the start marker and wait.” Confirm clearance before lowering. Suction remains on.
8. **Support and release:** At the verified low stationary pose: “You may approach and support the panel now. Tell me when you have it supported.” Release only after the physical confirmation. Then: “This trial is finished. Please wait while we save the recordings.”

Interruptions

If STOP is called, stop the activity using the established lab procedure, retain the attempt and record why it ended. Do not restart automatically. A fault-related repeat uses the same assigned controller/scenario slot and a new attempt ID under the existing repeat rule.

Use: “We paused because of a recording or equipment issue” only when that is the actual reason. Avoid “That controller failed”, “That one was safer” or “You should trust it”. Do not ask for safety ratings between trials.

Operator script: feedback and finish

Team only | Complete forms before revealing the comparison | v1.1

After each three-trial block

Read the current block letter in place of “A”:

> That completes Block A. Please scan this QR code, check your participant code and select Block A. Answer about all three trials in that block. There are no right or wrong answers. Please complete it privately; ask us if you need a question explained.

Confirm the matching block response arrived before proceeding. The same block form is used three times. Its participant code is prefilled; the participant currently selects A, B or C themselves. Offer a break between blocks and record breaks consistently.

If asked which condition is better: “We are comparing them and have not established a result. Please judge the experience you had.” If asked which controller is running: “We will explain the mapping after your final feedback.” Answer safety questions honestly; do not withhold necessary information to preserve blinding.

If wording is unclear, repeat the question and its scale neutrally. Do not suggest a score. Log any clarification, ambiguity or inability to answer. Do not present the adapted questionnaire as a full validated workload instrument.

Final form, then debrief

> You have completed Blocks A, B and C. Please scan the final QR code and check your participant code. Think back across all three blocks and answer based on your own experience. Let us know about any discomfort, fatigue or concern.

Wait for the final response before giving the following explanation:

> Thank you. We compared three ways of controlling robot speed and stopping: fixed distance zones, a response based on current separation and motion, and a response that also uses predicted motion and task context. We wanted to compare both the robot's response and how the task felt to you. We have not established which approach performed best.
>
> Each block included a waiting trial and two demonstrated movement trials. Keeping the task and movements consistent helps us compare the controllers. The results apply to this study setup and these movements. Do you have any remaining questions or concerns?

If asked, read this participant's actual controller mapping from the dashboard; A, B and C do not have one universal mapping. Refer to the approved study information for contact details and any data-withdrawal procedure. Do not invent a deadline. Confirm any concern has been handled, help remove the equipment, and verify files before dismissing the participant.

Team action card: complete once for the session

Use the released physical procedure. These are factual handover fields, not a new approval form. Brief every operator from the same completed card. If a detail is unknown, it remains unresolved; this pack does not fill it by assumption.

- Session date / release / lead: ___________________________________
- Robot operator / data operator: __________________________________
- Start marker / waiting area / stop instruction: ______________________
- Panel task: exact actions and completion rule: ______________________
- Non-closing action: path / pace / endpoint / return: __________________
- Closing action: path / pace / limit / immediate return: ________________
- Rehearsal and break procedure: __________________________________

Materials: give out the participant PDF; keep this script at the console. Use dashboard QR codes to preserve ID prefilling. Existing university study information and consent remain the source for participant rights and data handling. Operational release requirements remain in the team guide.